

Irreducible Max-Plus Powers: Exact Cyclicity, CSR Phases, Orbit Strata, and the Sharp Transient Boundary

Route-A structural certificate C188

Abstract

For every irreducible rational max-plus matrix A , normalize by its maximum cycle mean and let γ be the lcm of the cyclicities of the critical components. The source-locked cyclicity theorem says that γ is the minimal ultimate period of the entire matrix-power sequence. We integrate this with the exact first-equality transient, ultimate CS^tR representation, raw and projective orbit strata, attraction cones, the eigencone, and periodic ultimate spans. A fixed-support 2×2 family has every positive transient, while reducible matrices may have several growth rates. The classification is exact but has no intrinsic rational-prime semantics, target divisor, or Hilbert-space quantization.

Keywords: max-plus algebra; cyclicity; critical graph; CSR; attraction cone; transient.

中文摘要

本文对任意不可约有理max-plus矩阵给出统一的最终动力学分类。减去最大循环均值后，临界图各强连通分量循环性的最小公倍数 γ ，正是整个矩阵幂序列的最小最终周期。本文将这一有明确来源的经典定理与首次等式暂态、CSR表示、向量及射影轨道分层、吸引锥、特征锥和最终列张成空间联系起来。固定的二阶支撑可以实现任意大的暂态；可约系统则可能具有多个增长率。这些精确热带结构不产生内生的有理素数语义、目标除子或希尔伯特空间量子化。

关键词：max-plus代数；循环性；临界图；CSR；吸引锥；暂态。

1 Frozen system and classical input

On $\mathbb{Q}_{\max} = \mathbb{Q} \cup \{-\infty\}$ write $u \oplus v = \max(u, v)$ and $u \otimes v = u + v$. Let $A \in \mathbb{Q}_{\max}^{n \times n}$ have strongly connected finite support. Its maximum cycle mean is $\lambda(A)$, and $B = A - \lambda(A)$ means that $\lambda(A)$ is subtracted from every finite entry. The cited sources primarily use the equivalent max-times convention; the logarithm preserves powers, support, critical cycles and cyclicity while converting it exactly to this max-plus notation. The critical graph is the union of all cycles attaining $\lambda(A)$. If its strongly connected components have cyclicities $\gamma_1, \dots, \gamma_s$ —the gcds of their cycle lengths—put

$$\gamma = \text{lcm}(\gamma_1, \dots, \gamma_s).$$

The classical cyclicity theorem, in the convention source-locked to Sergeev [1], states more than a period bound: the *least* ultimate period of $\{B^t\}_{t \geq 0}$ is exactly γ . Thus there is a finite T_0 such that

$$B^{t+\gamma} = B^t \quad (t \geq T_0), \quad (1)$$

and no smaller positive integer is an ultimate matrix period. When the critical graph has several components the lcm, not a gcd across components, is essential.

The second source lock is the ultimate CSR theory of Sergeev and Schneider [2]. Put

$$K = (B^\gamma)^* = I \oplus B^\gamma \oplus \dots \oplus B^{(n-1)\gamma}.$$

Let C retain the critical columns of K , R its critical rows, and S the critical edges of B , filling all other entries by $-\infty$. For a matrix-dependent finite T_{CSR} ,

$$B^t = C \otimes S^t \otimes R \quad (t \geq T_{\text{CSR}}). \quad (2)$$

Neither source asserts a weight-independent uniform transient, and neither do we.

2 Complete eventual dynamics

Exact transient. Define $T(A)$ to be the least onset in (1). It has the sharp one-equality characterization

$$T(A) = \min\{t \geq 0 : B^{t+\gamma} = B^t\}. \quad (3)$$

Indeed, the classical theorem makes the set nonempty. If equality holds at one t , right multiplication by B^k gives $B^{t+k+\gamma} = B^{t+k}$ for every $k \geq 0$, proving (3).

Vectors, projective classes, and attraction cones. For every $x \in \mathbb{Q}_{\max}^n$, equation (1) shows that the ultimate period of $B^t \otimes x$ divides γ . For a nonzero vector, max-plus homogeneity gives the same conclusion after projectivization. Both periods may be smaller: for

$$B = \begin{pmatrix} -\infty & 0 \\ 0 & -\infty \end{pmatrix}, \quad x = (0, 0)^T,$$

the matrix period is two whereas x is fixed.

For $p \mid \gamma$, the exact attraction cone is

$$\text{Attr}_p(B) = \{x : B^{T+p} \otimes x = B^T \otimes x\}.$$

The same propagation proof shows that it consists precisely of vectors whose ultimate period divides p . Hence the exact-period- p stratum is

$$\text{Attr}_p(B) \setminus \bigcup_{\substack{q|p \\ q < p}} \text{Attr}_q(B).$$

Replacing equality by projective equality gives the projective strata. These are exact two-sided max-plus linear tests, not labels inferred from sampled orbits.

Eigencone and ultimate spans. The normalized eigencone $E(B) = \{x : B \otimes x = x\}$ lies in $\text{Attr}_1(B)$. For $0 \leq r < \gamma$, put $V_r = \text{Col}(B^{T+r})$. Then

$$V_{r+\gamma} = V_r, \quad B(V_r) = V_{r+1},$$

and every orbit lies in the corresponding phase cone after time T . We do not assert that all V_r are distinct.

3 Sharp boundaries

Primitive does not mean immediate. If $\gamma = 1$, the matrix powers are eventually constant. Nevertheless, for every integer $m \geq 1$ the fixed-support irreducible family

$$B_m = \begin{pmatrix} 0 & -m \\ 0 & -1 \end{pmatrix}$$

has critical cyclicity one and, for $t \geq 1$, direct induction gives

$$B_m^t = \begin{pmatrix} 0 & -m \\ 0 & \max(-t, -m) \end{pmatrix}. \quad (4)$$

Thus its minimal transient is exactly m . Fixed dimension and fixed support therefore admit unbounded transient as the weights vary; a dimension-only, weight-independent bound is false.

Reducibility changes the theorem. The reducible matrix $D = \text{diag}(0, 1)$ has component growth rates zero and one. After normalization by the larger rate, its powers contain $-t$ and are not a single periodic sequence. General reducible behavior is described by multiple growth rates and CSR terms [2]; we do not extend the irreducible conclusion across this boundary.

4 Exact evidence and Route-A stop

The exact ledger covers 177 matrices, 901 raw/projective vector rows, 441 simple cycles and 189 critical components. It checks 5,469 CSR cells and 7,471 propagated-period cells. An independent Karp/closure/Tarjan/gcd and binary-power checker passes 7,924 assertions. A separate SymPy path passes 10,615 checks; replay is byte exact; 137 repaired-hash and one stale-hash attacks are rejected. Among the regression vectors, 28 raw and 28 projective orbits have period strictly below their matrix period.

These checks are regression, not a finite proof of the source theorems. The package contribution is the source-locked synthesis, elementary derived classification and executable boundary certificate. Rational weights, support cycles and critical means carry no intrinsic rational-prime origin; ultimate phases and CSR identify no target divisor; no target functional equation, continuation, counting law or Weil compression follows; and a max-plus semimodule is not a source-native complex Hilbert-space quantization. Therefore

$$(A0, A1, A2, A3, A4) = (\text{FAIL}, \text{WEAK}, \text{FAIL}, \text{FAIL}, \text{FAIL}),$$

overall `ROUTE_A_REJECTED`, Route B false, with scope `NO_BAD_EULER_OR_ROOT_NUMBER`.

Ownership and derivation ledger. The equality of the minimal ultimate matrix period with γ is classical and owned by the first source; the ultimate CSR representation is owned by the second. The package-level increment is their convention-consistent synthesis with the one-equality transient proof, exact divisor strata, phase-indexed ultimate spans, the fixed-support sharp family, and the reducible firewall. The census validates these formulas on declared finite inputs but supplies no priority or completeness claim about the wider literature.

Revision-round focus. Round 2 separates classical ownership from finite regression, closes the reducible multirate boundary, and locks the semantic mutation audit and five-gate Route-A stop.

Limitations and nonclaims. We claim no priority for cyclicity, attraction-cone, ultimate-span, or CSR theory; no uniform transient; and no equality of every vector period with γ . We assert no arithmetic local data, target divisor, Hilbert-Pólya operator, global novelty, external review, or acceptance score.

Declarations

Data and code availability. Package-local exact evidence and deterministic code accompany this manuscript.

Ethics. No human, animal, clinical, personal, or sensitive data are used; approval is not applicable.

Author contributions (CRediT). This anonymous certificate records Conceptualization, Formal analysis, Software, Validation, Writing—original draft, and Writing—review and editing. AI systems are not authors.

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